

SWE_AGENT_ANALYSIS

SWE-agent Analysis

Tier 1 Agent - Agent for software engineering tasks **Source:** <https://github.com/princeton-nlp/SWE-agent> **Language:** Python **License:** MIT

Overview

SWE-agent is an open-source agent developed at Princeton NLP that performs software engineering tasks, particularly focused on GitHub issue resolution and bug fixing.

Core Features

1. SWE-bench Integration

- Evaluated on SWE-bench benchmark
- Real GitHub issue resolution
- Bug fixing capabilities

2. Computer Interface

- Filesystem navigation
- File viewing
- File editing
- Command execution

3. Thought-Action Loop

- Reasoning before action
- Action execution
- Observation feedback
- Iterative improvement

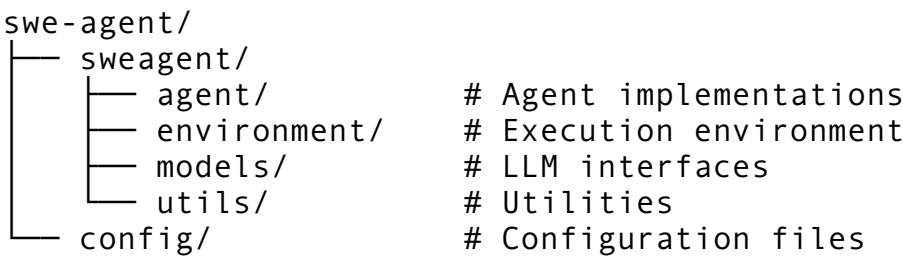
4. Demonstration Trajectories

- Training on demonstrations
- Few-shot learning
- Trajectory replay

5. Multi-Model Support

- GPT-4
- Claude
- Open-source models

Architecture



Key Capabilities

- 1. **Issue Resolution:** Fixes GitHub issues
- 2. **Code Navigation:** Navigates large codebases
- 3. **Bug Fixing:** Identifies and fixes bugs
- 4. **Evaluation:** SWE-bench performance

HelixAgent Integration Points

SWE-agent Feature	HelixAgent Implementation
Thought-Action	Agent workflow
Computer Interface	Tool system
Environment	Container runtime
Evaluation	Benchmark system

Documentation

- Website: <https://swe-agent.com>
- Paper: <https://arxiv.org/abs/2405.15793>
- GitHub: <https://github.com/princeton-nlp/SWE-agent>

Porting Priority: HIGH

SWE-agent’s issue resolution and evaluation framework are valuable additions.